

Ahan Bhatt

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EDUCATION

University of California, Santa Cruz [3.94 GPA] <i>Master of Science in Computer Science and Engineering</i>	Santa Cruz, California Aug. 2025 – Jun 2026
Indus Institute of Technology and Engineering [3.94 GPA] <i>Bachelor of Technology in Computer Engineering</i>	Ahmedabad, India Aug. 2020 – July 2024

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, R

Machine Learning: PyTorch, TensorFlow, Keras, Transformers, scikit-learn, NumPy, Pandas, Matplotlib

LLM and Data Tools: LangChain, ChromaDB, RAG, vector databases, PyPDF

Backend and Deployment: Node.js, FastAPI, Flask, REST APIs, Docker, Kubernetes, Linux, Git, CI/CD, Nginx

Cloud: AWS EC2, AWS S3, AWS IAM, AWS Lambda, SageMaker, CloudWatch

Data Engineering: PostgreSQL, Redis, ETL, SQL optimization, schema design, data cleaning, time-series preprocessing

Software Engineering: TDD, API integration, version control, DSA, System Design, Performance Optimization, Code Reviews, Debugging

EXPERIENCE

Student Researcher <i>AI Explainability and Accountability (AIEA) Lab @ UCSC</i>	September 2025 – December 2025 Santa Cruz, California
Designed and implemented a NeuroSymbolic AI framework for cognitive decline assessment from long-term AI usage, integrating neural pattern recognition with symbolic reasoning and fuzzy logic to produce interpretable risk scores and explainable reasoning traces.	
AI Research Intern <i>AI Institute of South Carolina</i>	June 2024 – May 2025 Columbia, South Carolina (remote)
Conducted research on multimodal toxicity mitigation and anomaly detection, using PostgreSQL-backed experiment tracking to manage dataset metadata, model outputs, evaluation metrics, and benchmarking results for projects like DE-HATE and Time Series Foundation Model evaluation.	
Data Science Research Trainee <i>Space Applications Centre, ISRO</i>	Jan. 2024 – May. 2024 Ahmedabad, Gujarat
Built an LSTM-based satellite clock bias prediction pipeline with automated time-series preprocessing and anomaly filtering, improving signal quality and enabling more accurate navigation timing predictions.	
Student Researcher (AI) <i>Indus Institute of Technology</i>	May 2022 – Jan. 2024 Ahmedabad, Gujarat
Developed Node.js-backed LLM pipelines for a university website chatbot, automated knowledge graph generation, and retrieval-augmented medical QA, evaluating GPT-4, LLaMA-2, and BERT for structured semantic extraction and grounded reasoning.	

PROJECTS

[Research Forge – A Self-Improving Research Agent](#)

- Working on Research Forge, a universal self-improving AI research agent that autonomously discovers papers, generates hypotheses, runs experiments, and updates its reasoning strategy using long-term graph memory.

[Meeting AI Assistant](#)

- Built a real-time AI meeting copilot desktop app (Electron + React + Express) that captures system audio, transcribes conversations, and streams context-aware responses using OpenAI API with screenshot and document context support.

[NeuroSymbolic Cognitive Decline Analyzer](#)

- Developed a hybrid NeuroSymbolic AI framework combining neural feature extraction with symbolic reasoning and fuzzy logic rules to assess cognitive decline risk associated with prolonged AI usage while maintaining interpretable reasoning traces.

Knowledge Graph Generation from Large Language Models

- Built an automated pipeline for generating structured knowledge graphs from unstructured text using large language models and evaluated GPT-4, LLaMA-2, and BERT for relation extraction and GraphRAG integration.

Time Series Foundation Model Evaluation for Anomaly Detection

- Benchmarked Time Series Foundation Models such as TimeGPT, Chronos, and Time-MOE across multiple datasets, showing that traditional statistical and deep learning models can outperform TSFMs for anomaly detection and prediction.

Med-Bot: Retrieval Augmented Medical Assistant

PUBLICATIONS

Leveraging LSTM for Predictive Modeling of Satellite Clock Bias

- *Presented at [CDMA 2025 conference](#) and available on [IEEE Xplore](#)*

Generating Knowledge Graphs from Large Language Models

Time Series Foundational Models: Their Role in Anomaly Detection and Prediction

- *Presented at [AAAI 2025 Anomaly Detection in Scientific Domains Workshop](#)*

Med-Bot: An AI-Powered Assistant to Provide Accurate and Reliable Medical Information